

# Homework 2

## Problem 1

(1)证明:

$$\binom{r}{r} + \binom{r+1}{r} + \binom{r+2}{r} + \cdots + \binom{n}{r} = \binom{n+1}{r+1}$$

(2)证明:

$$\sum_{k=0}^n \binom{m+k-1}{k} = \binom{n+m}{n}$$

**Answer:**

(1)

$$\binom{r}{r} = \binom{r+1}{r+1} = 1$$

根据

$$\binom{k+1}{l+1} = \binom{k}{l} + \binom{k}{l+1}$$

有

$$\begin{aligned} & \binom{r}{r} + \binom{r+1}{r} + \binom{r+2}{r} + \cdots + \binom{n}{r} \\ &= \binom{r+1}{r+1} + \binom{r+1}{r} + \binom{r+2}{r} + \cdots + \binom{n}{r} \\ &= \binom{r+2}{r+1} + \binom{r+2}{r} + \cdots + \binom{n}{r} \\ &= \cdots \\ &= \binom{n+1}{r+1} \end{aligned}$$

(2)

$$\begin{aligned}
\sum_{k=0}^n \binom{m+k-1}{k} &= \binom{m-1}{0} + \binom{m}{1} + \binom{m+1}{2} + \cdots + \binom{m+n-1}{n} \\
&= \binom{m}{0} + \binom{m}{1} + \binom{m+1}{2} + \cdots + \binom{m+n-1}{n} \\
&= \binom{m+1}{1} + \binom{m+1}{2} + \cdots + \binom{m+n-1}{n} \\
&= \cdots \\
&= \binom{n+m}{m}
\end{aligned}$$

## Problem 2

(1)求:  $\sum_{i=1}^n i(i-1)$ ,  $\sum_{i=1}^n i^2$

Hint: Assume Problem 1  $r=2$

(2)求:  $\sum_{i=1}^n i^3$

**Answer:**

(1)

$$\binom{2}{2} + \binom{3}{2} + \binom{4}{2} + \cdots + \binom{n}{2} = \binom{n+1}{3} = \frac{(n-1)n(n+1)}{6}$$

$$\binom{1}{1} + \binom{2}{1} + \binom{3}{1} + \cdots + \binom{n}{1} = \binom{n+1}{2} = \frac{n(n+1)}{2}$$

$$\therefore \sum_{i=1}^n i(i-1) = 2 \left( \binom{2}{2} + \binom{3}{2} + \binom{4}{2} + \cdots + \binom{n}{2} \right) = \frac{(n-1)n(n+1)}{3}$$

$$\therefore \sum_{i=1}^n i^2 = \sum_{i=1}^n i(i-1) + \sum_{i=1}^n i = \frac{(n-1)n(n+1)}{3} + \frac{n(n+1)}{2} = \frac{n(n+1)(2n+1)}{6}$$

(2)

$$\binom{3}{3} + \binom{4}{3} + \binom{5}{3} + \cdots + \binom{n}{3} = \binom{n+1}{4} = \frac{(n-2)(n-1)n(n+1)}{24}$$

$$\therefore \sum_{i=3}^n i(i-1)(i-2) = 6 \left( \binom{3}{3} + \binom{4}{3} + \binom{5}{3} + \cdots + \binom{n}{3} \right) = \frac{(n-2)(n-1)n(n+1)}{4}$$

$$\sum_{i=3}^n i(i-1)(i-2) = \sum_{i=1}^n i(i-1)(i-2) = \sum_{i=1}^n i^3 + \sum_{i=1}^n (-3i^2) + \sum_{i=1}^n 2i$$

$$\therefore \sum_{i=1}^n i^3 = \frac{(n-2)(n-1)n(n+1)}{4} + 3\frac{n(n+1)(2n+1)}{6} - 2\frac{n(n+1)}{2} = \frac{n^2(n+1)^2}{4}$$

## Problem 3

计算以下表达式，要求结果不能包含求和符号

$$(1) \sum_{k=1}^n \frac{1}{k} \binom{k}{m}$$

$$(2) \sum_{k=0}^n k \binom{k}{m}$$

**Answer:**

(1)

$$\frac{1}{k} \binom{k}{m} = \frac{1}{k} \frac{k!}{(k-m)!m!} = \frac{(k-1)!}{(k-m)!m!} = \frac{(k-1)!}{m(k-m)!(m-1)!} = \frac{1}{m} \binom{k-1}{m-1}$$

$$\begin{aligned} \sum_{k=1}^n \frac{1}{k} \binom{k}{m} &= \frac{1}{m} \sum_{k=1}^n \binom{k-1}{m-1} \\ &= \frac{1}{m} \left( \binom{0}{m-1} + \binom{1}{m-1} + \binom{2}{m-1} + \cdots + \binom{n-1}{m-1} \right) \\ &= \frac{1}{m} \left( \binom{m-1}{m-1} + \binom{m}{m-1} + \binom{m+1}{m-1} + \cdots + \binom{n-1}{m-1} \right) \\ &= \frac{1}{m} \binom{n}{m} \end{aligned}$$

(2)

$$k \binom{k}{m} = (k+1) \binom{k}{m} - \binom{k}{m}$$

$$(k+1) \binom{k}{m} = \frac{(k+1)k!}{(k-m)!m!} = \frac{(k+1)!}{(k-m)!m!} = \frac{(m+1)(k+1)!}{(k-m)!(m+1)!} = (m+1) \binom{k+1}{m+1}$$

$$\begin{aligned}
\sum_{k=0}^n k \binom{k}{m} &= (m+1) \sum_{k=0}^n \binom{k+1}{m+1} - \sum_{k=0}^n \binom{k}{m} \\
&= (m+1) \left( \binom{1}{m+1} + \binom{2}{m+1} + \binom{3}{m+1} + \cdots + \binom{n+1}{m+1} \right) \\
&\quad - \left( \binom{0}{m} + \binom{1}{m} + \binom{2}{m} + \cdots + \binom{n}{m} \right) \\
&= (m+1) \left( \binom{m+1}{m+1} + \binom{m+2}{m+1} + \binom{m+3}{m+1} + \cdots + \binom{n+1}{m+1} \right) \\
&\quad - \left( \binom{m}{m} + \binom{m+1}{m} + \binom{m+2}{m} + \cdots + \binom{n}{m} \right) \\
&= (m+1) \binom{n+2}{m+2} - \binom{n+1}{m+1}
\end{aligned}$$

## Problem 4

求  $f: \{1, 2, 3, \dots, n\} \rightarrow \{1, 2, 3, \dots, n\}$  中非严格增函数的个数

**Answer:**

所求函数的个数与  $g: \{0, 1, 2, \dots, n+1\} \rightarrow \{1, 2, 3, \dots, n\}$ ,  $g(0) = 1, g(n+1) = n$  且  $g$  为非严格增函数的个数相同

令  $k_i = g(i+1) - g(i)$ , 则有  $\sum_{i=0}^n k_i = n - 1$

所以有  $g$  的个数为  $k_0 + k_1 + k_2 + \cdots + k_n$  的非负整数解个数

用隔板法易知  $g$  的个数为  $\binom{2n-1}{n}$